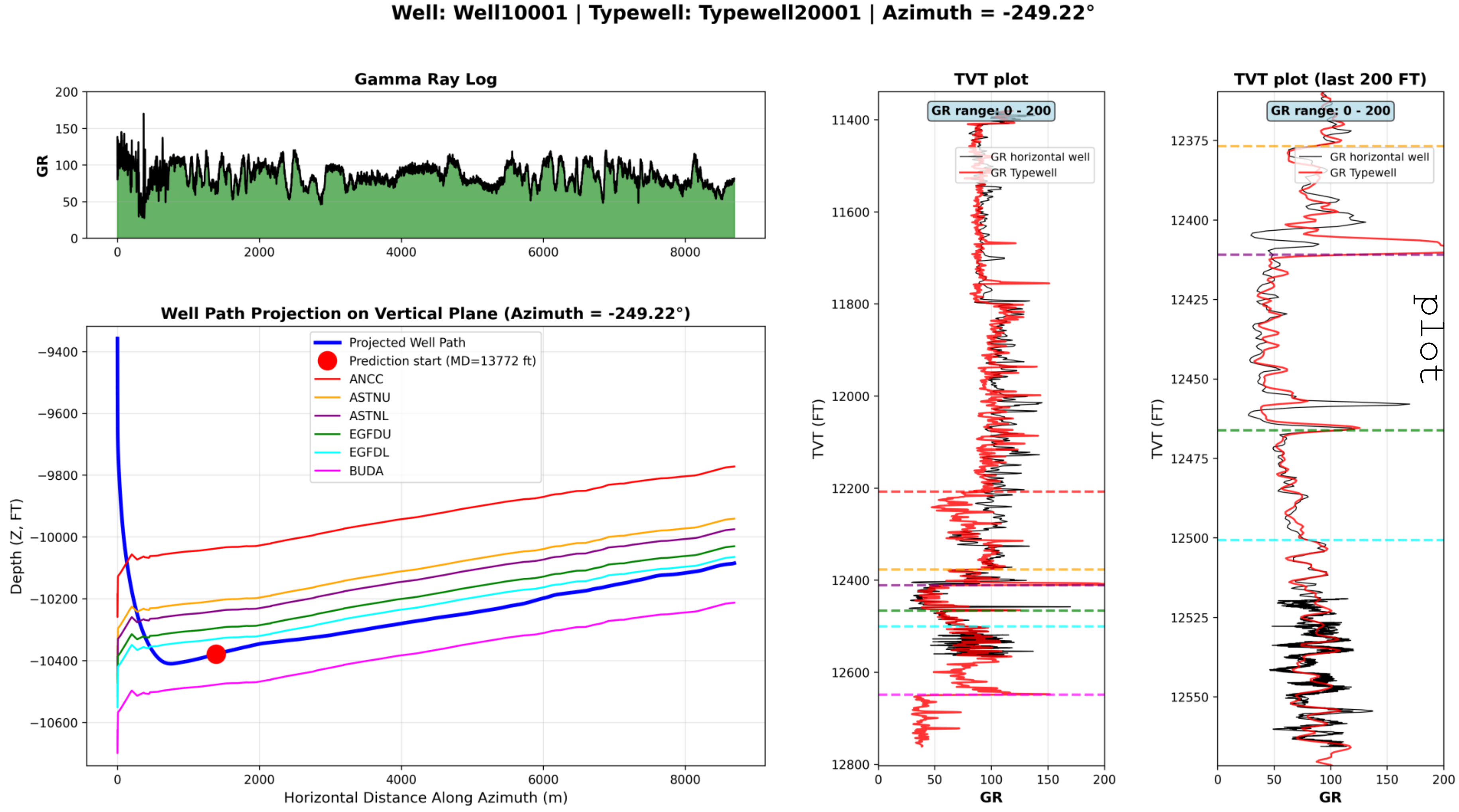
Wellbore Geology Prediction

Data Available



Zoomed in TVT plot

Each well has two CSV files:
Well1XXXX__horizontal_well.csv – horizontal well data
Well1XXXX__typewell__Typewell2XXXX.csv – Typewell (vertical well data)

Horizontal Well Data

Well10001__horizontal_well.csv file

	A	B	C	D	E	F	G	H	I	J	K	L	M	N
1	WELLNAME	MD	X	Y	Z	ANCC	ASTNU	ASTNL	EGFDU	EGFDL	BUDA	TVT	GR	TVT_input
2108	Well10001	#####	3010546.4	1076195.8	-10380.5	-10037.9	-10206.7	-10240.9	-10296.1	-10330.6	-10478.6	12550.6	74.3617565	12550.6
2109	Well10001	#####	3010546.1	1076196.7	-10380.5	-10037.9	-10206.7	-10240.8	-10296	-10330.6	-10478.6	12550.6	73.0787159	12550.6
2110	Well10001	#####	3010545.7	1076197.6	-10380.4	-10037.9	-10206.6	-10240.8	-10296	-10330.6	-10478.5	12550.6	69.7573697	12550.6
2111	Well10001	#####	3010545.3	1076198.6	-10380.3	-10037.9	-10206.6	-10240.8	-10296	-10330.6	-10478.5	12550.5	65.8991484	12550.5
2112	Well10001	#####	3010545	1076199.5	-10380.3	-10037.8	-10206.6	-10240.8	-10295.9	-10330.5	-10478.5	12550.5	61.4221557	12550.5
2113	Well10001	#####	3010544.6	1076200.4	-10380.2	-10037.8	-10206.5	-10240.7	-10295.9	-10330.5	-10478.4	12550.5	60.1209159	12550.5
2114	Well10001	#####	3010544.3	1076201.4	-10380.2	-10037.8	-10206.5	-10240.7	-10295.9	-10330.5	-10478.4	12550.4	61.11277	12550.4
2115	Well10001	#####	3010543.9	1076202.3	-10380.1	-10037.7	-10206.5	-10240.7	-10295.9	-10330.4	-10478.4	12550.4		12550.4
2116	Well10001	#####	3010543.5	1076203.2	-10380	-10037.7	-10206.4	-10240.6	-10295.8	-10330.4	-10478.4	12550.4	63.1510756	12550.4
2117	Well10001	#####	3010543.2	1076204.1	-10380	-10037.7	-10206.4	-10240.6	-10295.8	-10330.4	-10478.3	12550.3	62.9963828	12550.3
2118	Well10001	#####	3010542.8	1076205.1	-10379.9	-10037.6	-10206.4	-10240.6	-10295.8	-10330.3	-10478.3	12550.3	62.5141051	12550.3
2119	Well10001	#####	3010542.4	1076206	-10379.8	-10037.6	-10206.4	-10240.5	-10295.7	-10330.3	-10478.3	12550.3	62.0682258	12550.3
2120	Well10001	#####	3010542.1	1076206.9	-10379.8	-10037.6	-10206.3	-10240.5	-10295.7	-10330.3	-10478.2	12550.2	62.4049102	12550.2
2121	Well10001	#####	3010541.7	1076207.9	-10379.7	-10037.6	-10206.3	-10240.5	-10295.7	-10330.3	-10478.2	12550.2		12550.2
2122	Well10001	#####	3010541.3	1076208.8	-10379.7	-10037.5	-10206.3	-10240.5	-10295.6	-10330.2	-10478.2	12550.2	66.3632269	12550.2
2123	Well10001	#####	3010541	1076209.7	-10379.6	-10037.5	-10206.2	-10240.4	-10295.6	-10330.2	-10478.1	12550.1	66.9455999	
2124	Well10001	#####	3010540.6	1076210.7	-10379.5	-10037.5	-10206.2	-10240.4	-10295.6	-10330.2	-10478.1	12550.1	69.2932912	
2125	Well10001	#####	3010540.3	1076211.6	-10379.5	-10037.4	-10206.2	-10240.4	-10295.6	-10330.1	-10478.1	12550.1	75.7812907	
2126	Well10001	#####	3010539.9	1076212.5	-10379.4	-10037.4	-10206.1	-10240.3	-10295.5	-10330.1	-10478.1	12550	77.5011111	
2127	Well10001	#####	3010539.5	1076213.4	-10379.4	-10037.4	-10206.1	-10240.3	-10295.5	-10330.1	-10478	12550	76.4364604	

TVT (geology of the wellbore) must be predicted. This value is provided only for the training dataset

Top depth of each geological formation (provided only in the training dataset)

Provided for training and for prediction

Units: feet

MD – measured depth (well length)

XYZ – coordinates of each point of the well

GR – GR – gamma ray values measured at each point of the horizontal well (some values may be NaN)

TVT – geology values (see next slide)

TVT_input – same geology values available until the Prediction Start (PS) point

Vertical (Typewell) Well Data

Well10001__typewell__Typewell20001.csv file

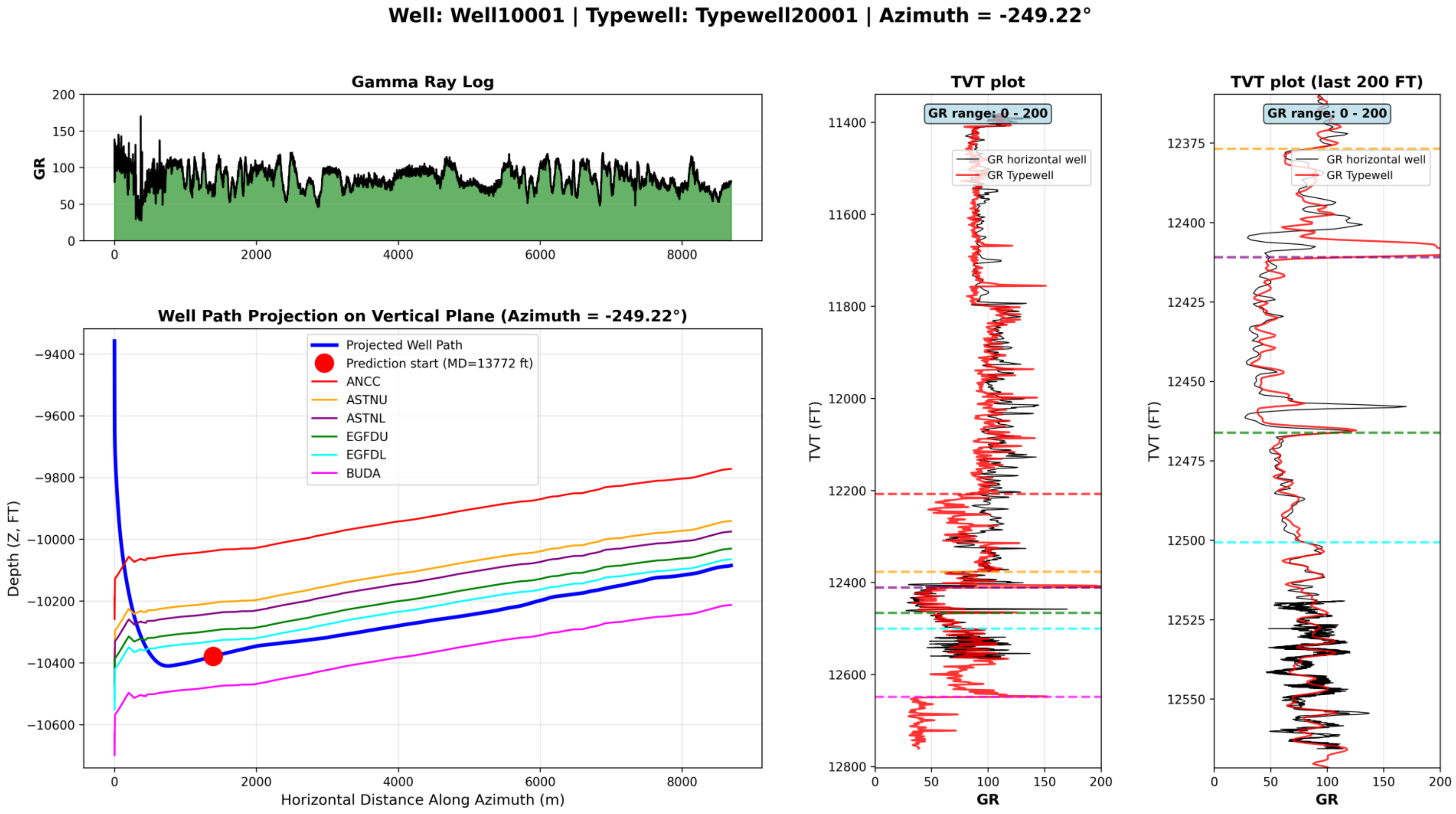
Typewell assigned to Horizontal well

1	TVT	GR	Geology
2187	12460.95	40.37	ASTNL
2188	12461.45	42.21	ASTNL
2189	12461.95	43.05	ASTNL
2190	12462.45	42.74	ASTNL
2191	12462.95	44.12	ASTNL
2192	12463.45	51.3	ASTNL
2193	12463.95	68.49	ASTNL
2194	12464.45	94.51	ASTNL
2195	12464.95	118.01	ASTNL
2196	12465.45	125.74	ASTNL
2197	12465.95	114.87	ASTNL
2198	12466.45	94.37	EGFDU
2199	12466.95	75.68	EGFDU
2200	12467.45	64.49	EGFDU
2201	12467.95	59.62	EGFDU
2202	12468.45	58.03	EGFDU
2203	12468.95	57.82	EGFDU
2204	12469.45	57.47	EGFDU
2205	12469.95	55.81	EGFDU
2206	12470.45	53.17	EGFDU
2207	12470.95	51.6	EGFDU
2208	12471.45	52.1	EGFDU
2209	12471.95	53.68	EGFDU
2210	12472.45	55.79	EGFDU
2211	12472.95	57.96	EGFDU
2212	12473.45	58.4	EGFDU
2213	12473.95	57.16	EGFDU
2214	12474.45	57.22	EGFDU
2215	12474.95	58.99	EGFDU
2216	12475.45	59.56	EGFDU
2217	12475.95	57.85	EGFDU
2218	12476.45	55.95	EGFDU
2219	12476.95	55.25	EGFDU
2220	12477.45	56.3	EGFDU
2221	12477.95	59.58	EGFDU
2222	12478.45	62.65	EGFDU

Each horizontal well is assigned one typewell

Units: feet
TVT - true vertical thickness (depth in the vertical well)
GR - Gamma Ray Values measured at each point of the vertical well
Geology - name of geological layer

Goal: get TVT values from GR data

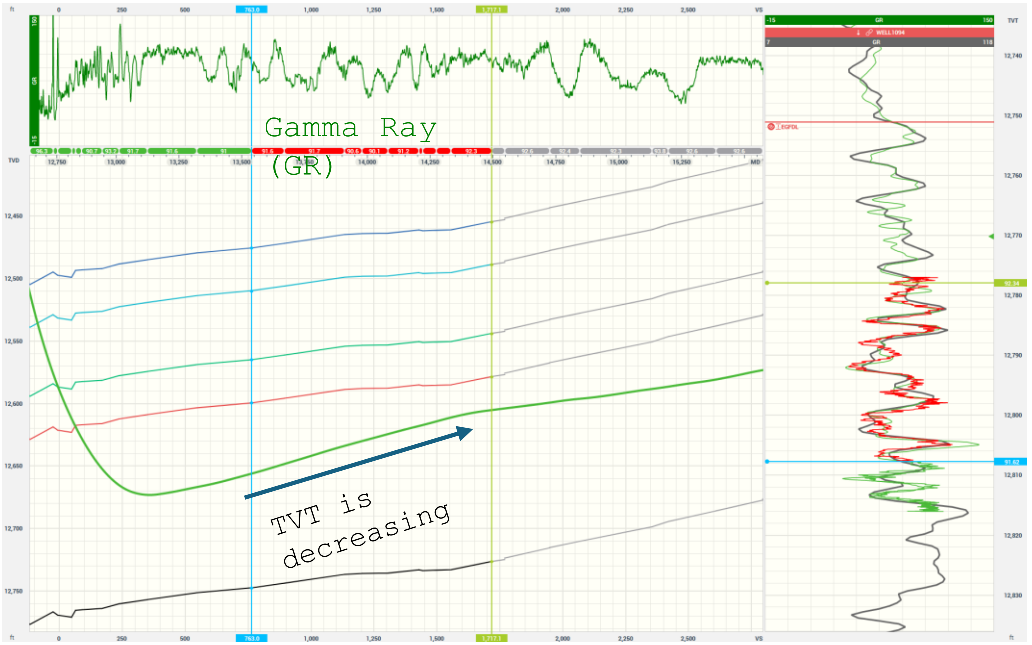


TVT value for typewell (vertical well) is always known

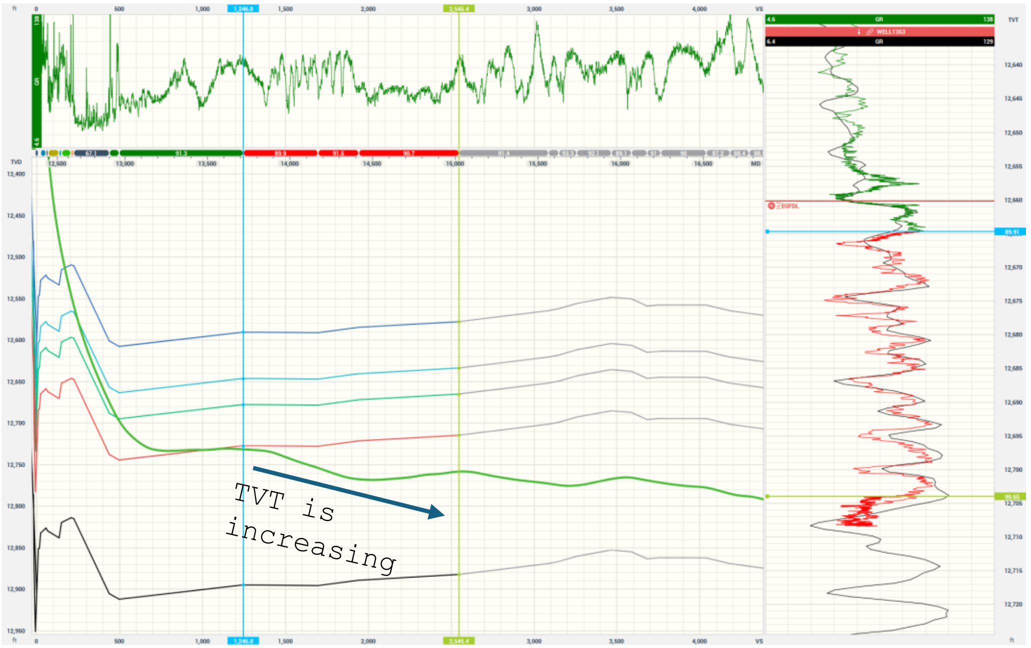
TVT values for the horizontal well are known until the Prediction Start (PS) point

The goal is to calculate TVT values beyond the PS point using XYZ and GR from the horizontal well and TVT and GR from the typewell

Goal: get TVT values from GR data

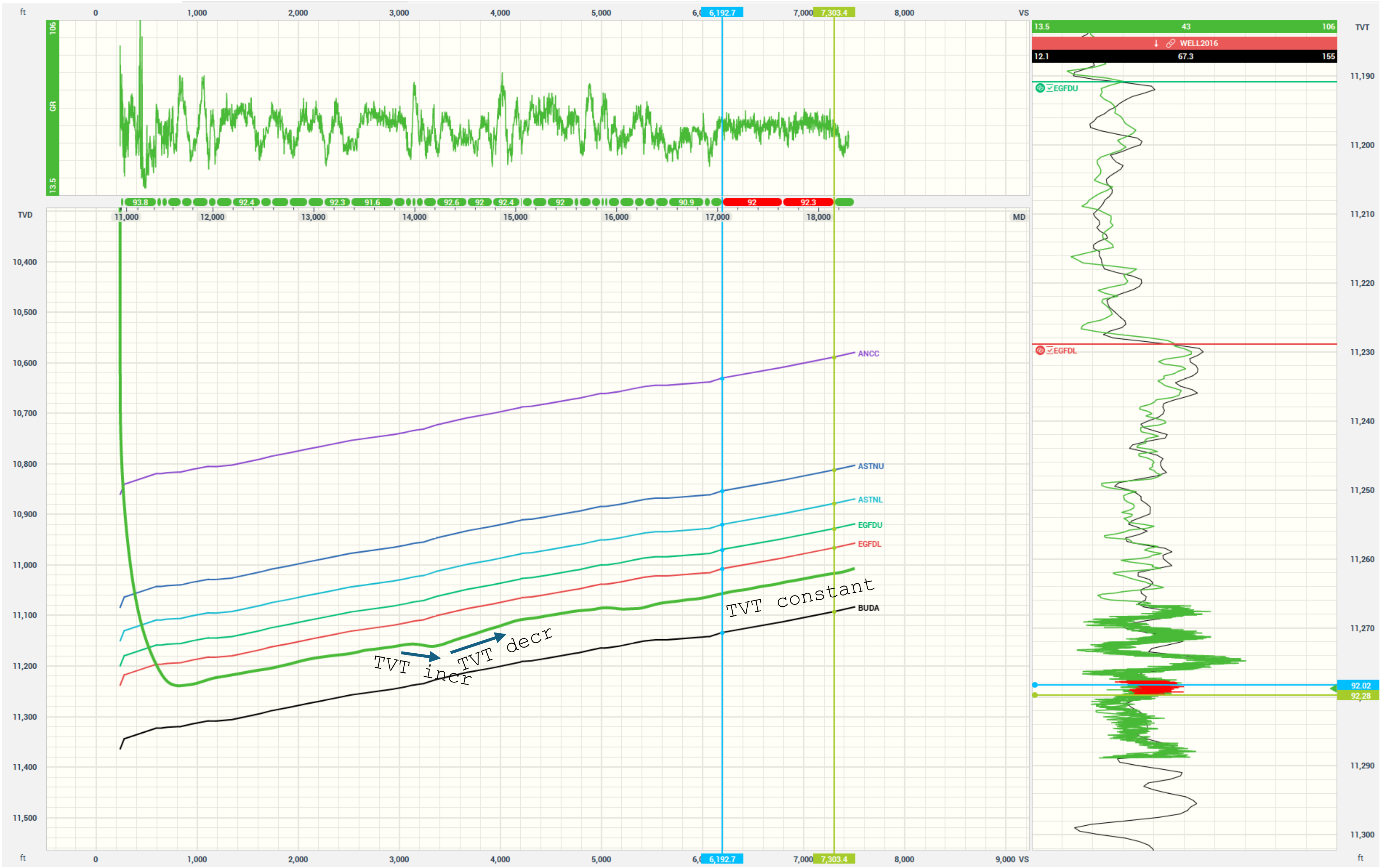


GR projected on TVT track
GR signature matches Typewell
GR
TVT is **decreasing**



GR projected on TVT track
GR signature matches Typewell
GR
TVT is **increasing**

Goal: get TVT values from GR data



GR projected on TVT track
GR signature is constant
TVT is **constant**

Understanding TVT from Gamma Ray data

TVT scale

VIDEO: GR behavior helps us understand the TVT (geology) along each portion of the horizontal well

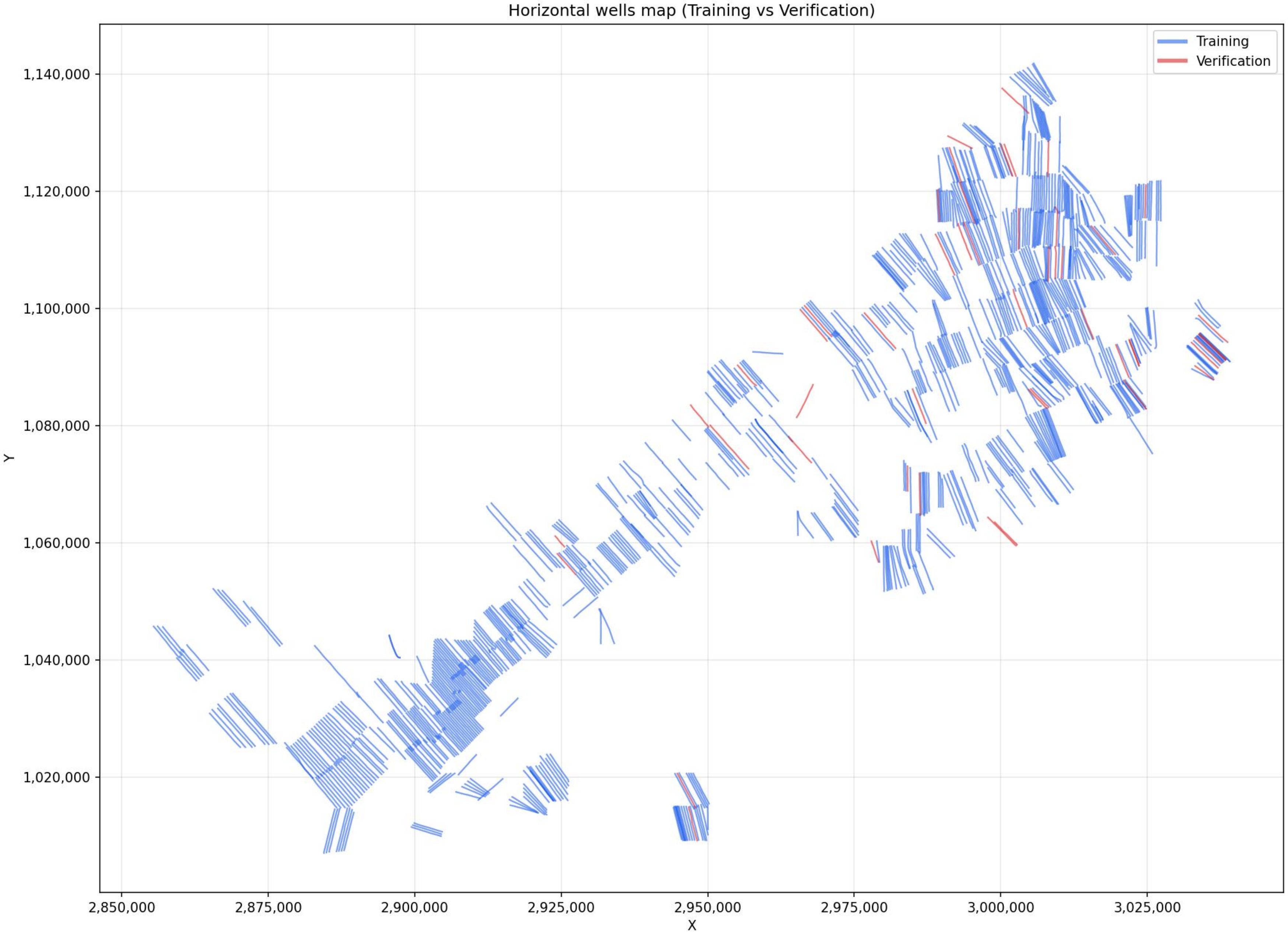
Gamma ray in the horizontal well has better resolution

GR from the vertical well has better resolution than GR from the typewell

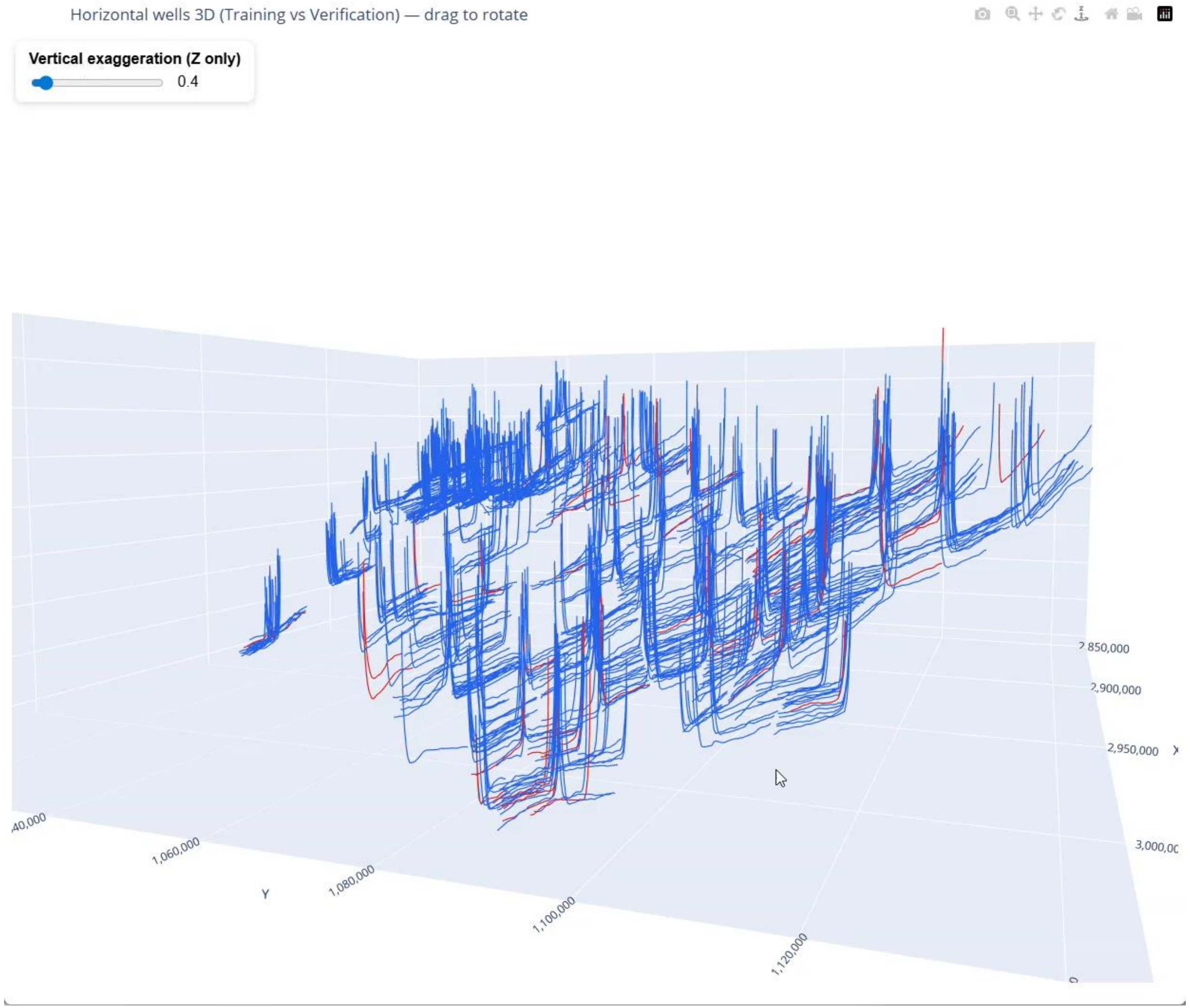
The green GR correlates better with the red GR than with the typewell GR (black)

It may be better to use GR data from the horizontal well before the PS point, combined with deeper TVT data, to correlate the rest of the lateral

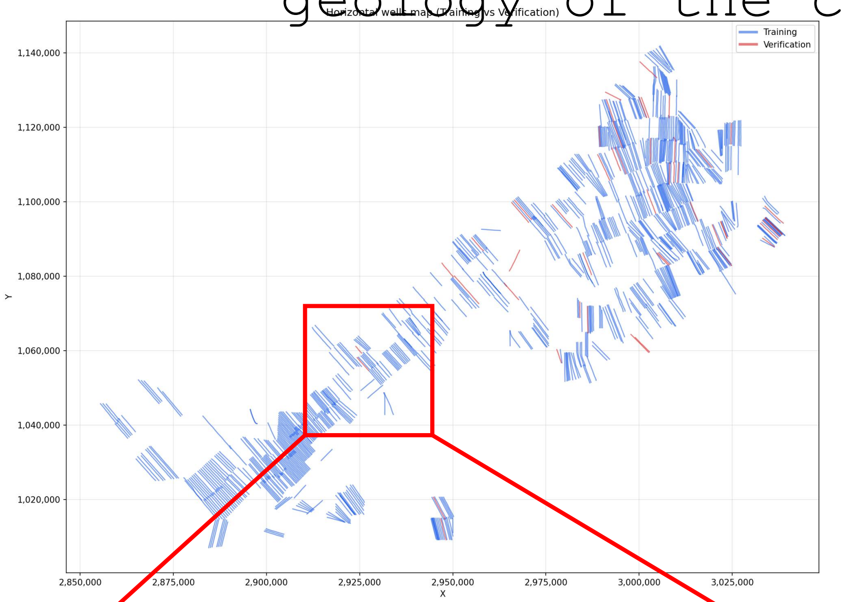
Map view of all training and validation wells

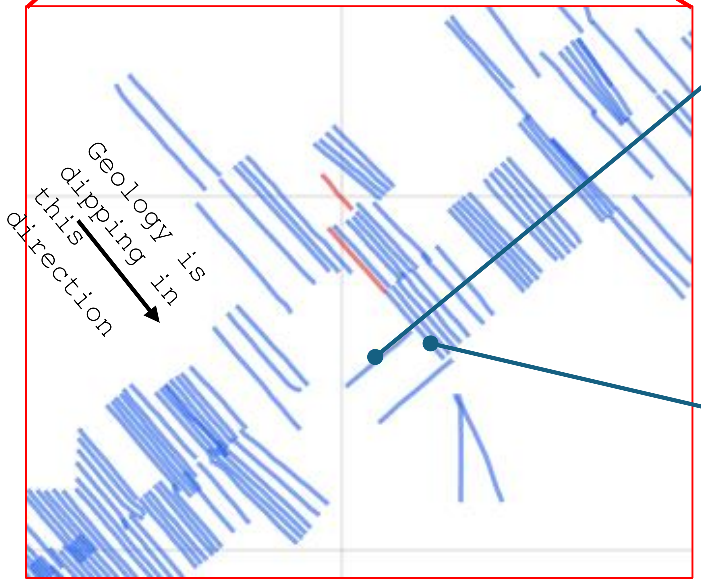


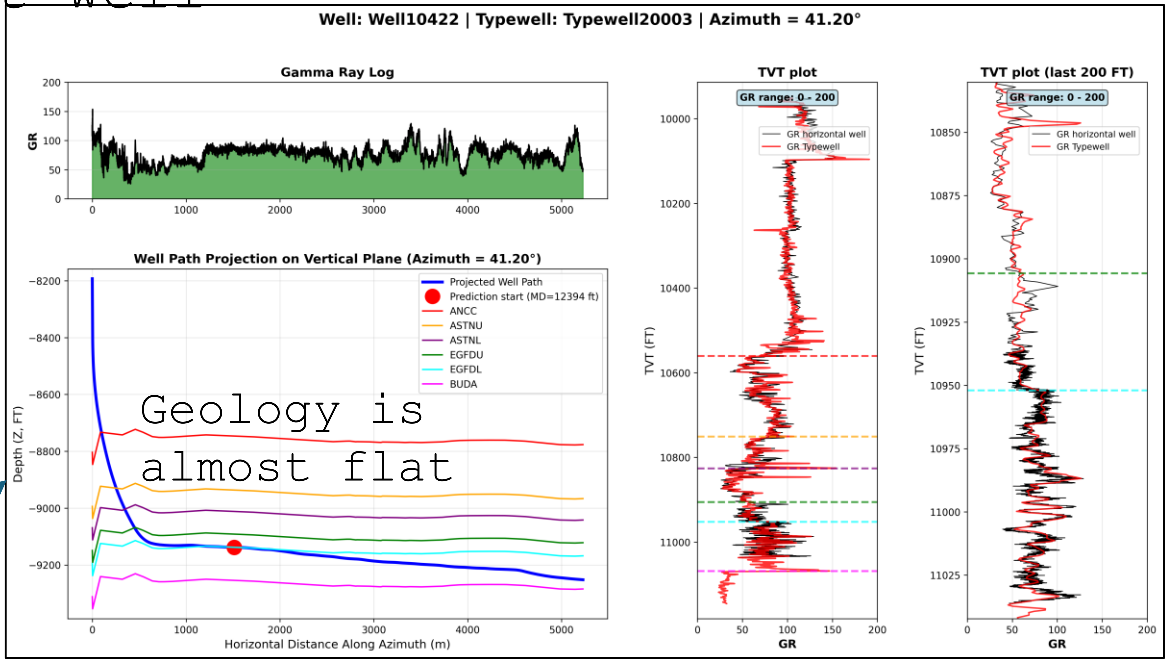
3D view of all training and validation wells

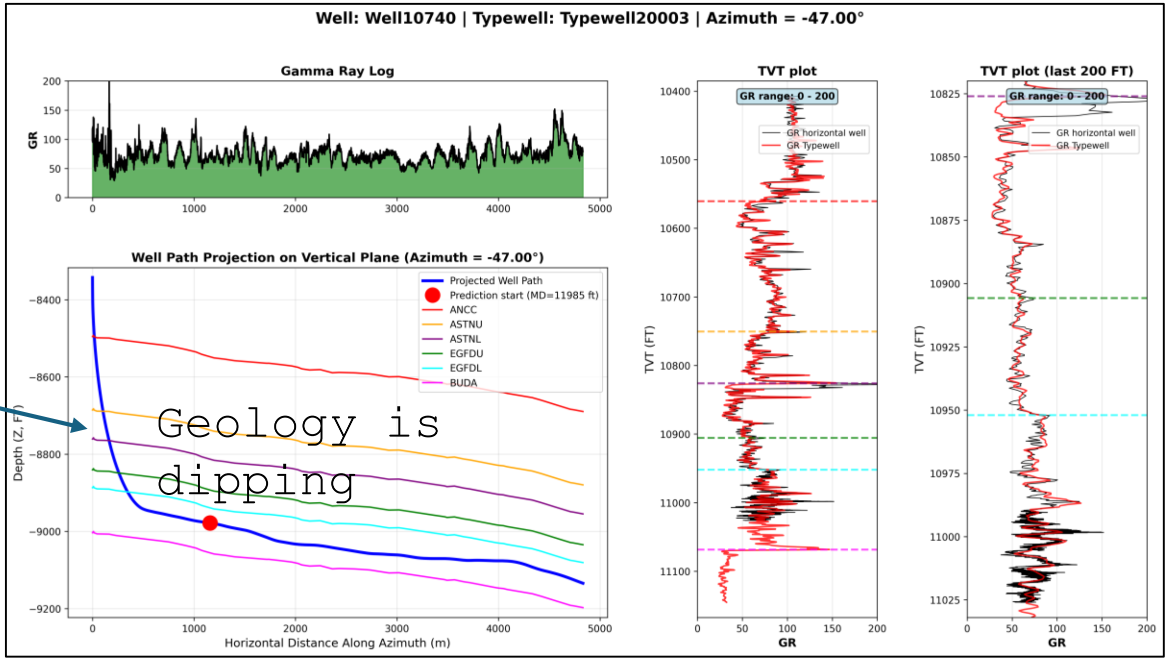


The geology of an offset well can help predict the geology of the current well



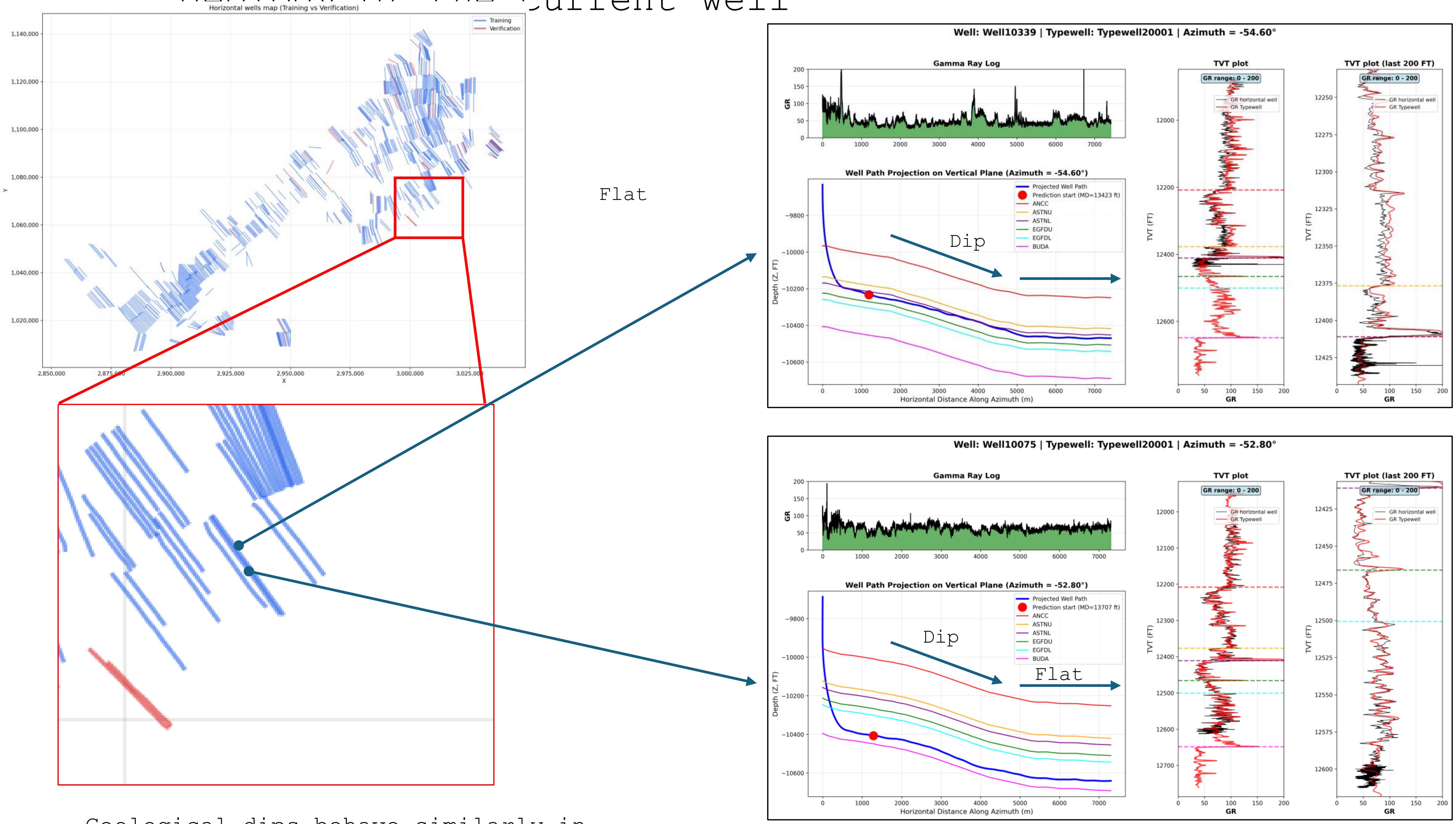






The azimuth of horizontal drilling affects the expected geology dip

The geology of an offset well can help predict the geology of the current well



Geological dips behave similarly in neighboring wells

How to evaluate Prediction quality

Each horizontal well has a TVT values that need to be predicted (one foot step)
dTVT = manualTVT-predictedTVT for each point that is predicted

Prediction quality is measured as the RMSE of all dTVT values